

Solving Exponential Equations with Common Bases

Matching bases, rewriting to a common base, and reading exponential tables

Directions

- Rewrite both sides as powers of the *same* base, then set the exponents equal.
- Show the common-base line before you set exponents equal — that line is the method.
- Give exact answers. Leave fractions as fractions; do not round.

1. Solve for x : $3^{x+2} = 3^5$

Answer: _____

2. Solve for x : $4^x = 64$

Answer: _____

3. The table gives values of $f(x) = 2^x$.

x	1	2	3	4
$f(x)$	2	4	8	16

- (a) Extend the pattern one row further: find $f(5)$.

Answer: _____

- (b) Write 256 as a power of 2 and solve $2^x = 256$.

Answer: _____

4. Solve for x : $5^{2x-1} = 5^{x+4}$

Answer: _____

5. Solve for x : $8^x = 32$

Answer: _____

6. A bacteria culture is modelled by $P(t) = 50 \cdot 2^t$, where P is the number of cells and t is the time in hours. The table shows the first four hours.

t (hours)	0	1	2	3
$P(t)$ (cells)	50	100	200	400

(a) How many cells are there after 5 hours? Find $P(5)$.

Answer: _____

(b) Solve $50 \cdot 2^t = 3200$ to find when the culture reaches 3200 cells.

Answer: _____

7. Solve for x : $9^{x-1} = 27^x$

Answer: _____

8. Solve for x : $7^{x^2} = 7^{3x+4}$

Answer: _____

9. The table gives values of $h(x) = 3^x$.

x	-1	0	1	2
$h(x)$	$\frac{1}{3}$	1	3	9

- (a) Continue the table to the left: find $h(-3)$.

Answer: _____

- (b) Solve $3^x = \frac{1}{81}$.

Answer: _____

10. Solve for x : $\left(\frac{1}{2}\right)^{x+1} = 8^{x-3}$

Answer: _____